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Course/Section:CPE31S1	**Date Submitted:01/15/2026**
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Activity 5: Consolidating Playbook plays

1. **Objectives:**

1.1 Use **when** command in playbook for different OS distributions

1.2 Apply refactoring techniques in cleaning up the playbook codes

2. **Discussion**:

We are going to look at a way that we can differentiate a playbook by a host in terms of which distribution the host is running. It's very common in most Linux shops to run multiple distributions, for example, Ubuntu shop or Debian shop and you need a different distribution for a one off-case or perhaps you want to run plays only on certain distributions.

It is a best practice in ansible when you are working in a collaborative environment to use the command git pull. git pull is a Git command used to update the local version of a repository from a remote. By default, git pull does two things. Updates the current local working branch (currently checked out branch) and updates the remote-tracking branches for all other branches. git pull essentially pulls down any changes that may have happened since the last time you worked on the repository.

Requirement:

In this activity, you will need to create a CentOS VM. Likewise, you need to activate the second adapter to a host-only adapter after the installations. Take note of the IP address of the CentOS VM. Make sure to use the command *ssh-copy-id* to copy the public key to CentOS. Verify if you can successfully SSH to CentOS VM.

Task 1: Use when command for different distributions

1. In the local machine, make sure you are in the local repository directory (*CPE232_yourname*). Issue the command git pull. When prompted, enter the correct passphrase or password. Describe what happened when you issue this command. Did something happen? Why?

```
isaac@Workstation:~/CPE212_Alegre$ git pull
Already up to date.
isaac@Workstation:~/CPE212_Alegre$
```

Already up to date.

2. Edit the inventory file and add the IP address of the Centos VM. Issue the command we used to execute the playbook (the one we used in the last activity): *ansible-playbook --ask-become-pass install_apache.yml*. After executing this command, you may notice that it did not become successful in the Centos VM. You can see that the Centos VM has failed=1. Only the two remote servers have been changed. The reason is that Centos VM does not support "apt" as the package manager. The default package manager for Centos is "yum."

```
isaac@Workstation:~/ansible$ ansible-playbook install_apache.yml -K
BECOME password:

PLAY [all] *********************************************************************************

TASK [Gathering Facts] *********************************************************************
[WARNING]: Platform linux on host 192.168.56.112 is using the discovered Python interpreter at
/usr/bin/python3.10, but future installation of another Python interpreter could change the mean
that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [192.168.56.112]

TASK [update repository index] *************************************************************
changed: [192.168.56.112]

TASK [install apache2 package] *************************************************************
changed: [192.168.56.112]

TASK [install php package] *****************************************************************
changed: [192.168.56.112]

TASK [add php support for apache] **********************************************************
ok: [192.168.56.112]

PLAY RECAP *********************************************************************************
192.168.56.112             : ok=5    changed=3    unreachable=0    failed=0    skipped=0    resc
  ignored=0
```

3. Edit the *install_apache.yml* file and insert the lines shown below.

```yaml
---
- hosts: all
  become: true
  tasks:

  - name: update repository index
    apt:
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: install apache2 package
    apt:
      name: apache2
    when: ansible_distribution == "Ubuntu"

  - name: add PHP support for apache
    apt:
      name: libapache2-mod-php
    when: ansible_distribution == "Ubuntu"
```

install_apache.yml

```yaml
- hosts: all
  become: true
  tasks:

  - name: update repository index
    apt:
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: install apache2 package
    apt:
      name: apache2
    when: ansible_distribution == "Ubuntu"

  - name: install php package
    apt:
      name: libapache2-mod-php
    when: ansible_distribution == "Ubuntu"

  - name: add php support for apache
    apt:
      name: libapache2-mod-php
    when: ansible_distribution == "Ubuntu"
```

Make sure to save the file and exit.

Run *ansible-playbook --ask-become-pass install_apache.yml* and describe the result.

If you have a mix of Debian and Ubuntu servers, you can change the configuration of your playbook like this.

```
- name: update repository index
  apt:
    update_cache: yes
  when: ansible_distribution in [“Debian”, “Ubuntu]
```

Note: This will work also if you try. Notice the changes are highlighted.

4. Edit the *install_apache.yml* file and insert the lines shown below.

```
---
- hosts: all
  become: true
  tasks:

  - name: update repository index
    apt:
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: install apache2 package
    apt:
      name: apache2
      stae: latest
    when: ansible_distribution == "Ubuntu"

  - name: add PHP support for apache
    apt:
      name: libapache2-mod-php
      state: latest
    when: ansible_distribution == "Ubuntu"

  - name: update repository index
    dnf:
      update_cache: yes
    when: ansible_distribution == "CentOS"

  - name: install apache2 package
    dnf:
      name: httpd
      state: latest
    when: ansible_distribution == "CentOS"

  - name: add PHP support for apache
    dnf:
      name: php
      state: latest
    when: ansible_distribution == "CentOS"
```

Make sure to save and exit.

Run *ansible-playbook --ask-become-pass install_apache.yml* and describe the result.

```
isaac@Workstation:~/ansible$ ansible-playbook install_apache.yml -K
BECOME password:

PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
[WARNING]: Platform linux on host 192.168.56.112 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [192.168.56.112]

TASK [update repository index] *************************************************
changed: [192.168.56.112]

TASK [install apache2 package] *************************************************
ok: [192.168.56.112]

TASK [install php package] *****************************************************
ok: [192.168.56.112]

TASK [add php support for apache] **********************************************
ok: [192.168.56.112]

PLAY RECAP *********************************************************************
192.168.56.112             : ok=5    changed=1    unreachable=0    failed=0    skippe
    rescued=0    ignored=0
```

5. To verify the installations, go to CentOS VM and type its IP address on the browser. Was it successful? The answer is no. It's because the httpd service or the Apache HTTP server in the CentOS is not yet active. Thus, you need to activate it first.

5.1 To activate, go to the CentOS VM terminal and enter the following:
systemctl status httpd
The result of this command tells you that the service is inactive.

5.2 Issue the following command to start the service:
sudo systemctl start httpd
(When prompted, enter the sudo password)
sudo firewall-cmd --add-port=80/tcp
(The result should be a success)

5.3 To verify the service is already running, go to CentOS VM and type its IP address on the browser. Was it successful? (Screenshot the browser)

Task 2: Refactoring playbook

This time, we want to make sure that our playbook is efficient and that the codes are easier to read. This will also makes run ansible more quickly if it has to execute fewer tasks to do the same thing.

1. Edit the playbook *install_apache.yml*. Currently, we have three tasks targeting our Ubuntu machines and 3 tasks targeting our CentOS machine. Right now, we try to consolidate some tasks that are typically the same. For example, we can consolidate two plays that install packages. We can do that by creating a list of installation packages as shown below:

```
---
- hosts: all
  become: true
  tasks:

  - name: update repository index Ubuntu
    apt:
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: install apache2 and php packages for Ubuntu
    apt:
      name:
        - apache2
        - libapache2-mod-php
      state: latest
    when: ansible_distribution == "Ubuntu"

  - name: update repository index for CentOS
    dnf:
      update_cache: yes
    when: ansible_distribution == "CentOS"

  - name: install apache and php packages for CentOS
    dnf:
      name:
        - httpd
        - php
      state: latest
    when: ansible_distribution == "CentOS"
```

Make sure to save the file and exit.

playbook(first act).txt | install_apache.yml | inventory.ini | ansible.

install_apache.yml

```yaml
- hosts: all
  become: true
  tasks:

  - name: update repository index
    apt:
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: install apache2 package
    apt:
      name: apache2
      name: libapache2-mod-php
      state: latest
    when: ansible_distribution == "Ubuntu"
```

Run *ansible-playbook --ask-become-pass install_apache.yml* and describe the result.

2. Edit the playbook *install_apache.yml* again. In task 2.1, we consolidated the plays into one play. This time we can actually consolidated everything in just 2 plays. This can be done by removing the update repository play and putting the command *update_cache: yes* below the command *state: latest*. See below for reference:

```yaml
---
- hosts: all
  become: true
  tasks:

   - name: install apache2 and php packages for Ubuntu
     apt:
      name:
        - apache2
        - libapache2-mod-php
      state: latest
      update_cache: yes
     when: ansible_distribution == "Ubuntu"

   - name: install apache and php packages for CentOS
     dnf:
       name:
         - httpd
         - php
       state: latest
       update_cache: yes
     when: ansible_distribution == "CentOS"
```

Make sure to save the file and exit.

Run *ansible-playbook --ask-become-pass install_apache.yml* and describe the result.

playbook(first act).txt | install_apache.yml | inventory.ini | ansible.c

install_apache.yml

```yaml
- hosts: all
  become: true
  tasks:

  - name: install apache2 package
    apt:
      name: apache2
      name: libapache2-mod-php
      state: latest
      update_cache: yes
    when: ansible_distribution == "Ubuntu"
```

```
isaac@Workstation:~/ansible$ ansible-playbook install_apache.yml -K
BECOME password:
[WARNING]: While constructing a mapping from /home/isaac/ansible/install_apache.yml,
line 7, column 7, found a duplicate dict key (name). Using last defined value only.

PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
[WARNING]: Platform linux on host 192.168.56.112 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [192.168.56.112]

TASK [install apache2 package] *************************************************
ok: [192.168.56.112]

PLAY RECAP *********************************************************************
192.168.56.112             : ok=2    changed=0    unreachable=0    failed=0    skippe
    rescued=0    ignored=0
```

3. Finally, we can consolidate these 2 plays in just 1 play. This can be done by declaring variables that will represent the packages that we want to install. Basically, the apache_package and php_package are variables. The names are arbitrary, which means we can choose different names. We also take out the line when: ansible_distribution. Edit the playbook *install_apache.yml* again and make sure to follow the below image. Make sure to save the file and exit.

```
---
- hosts: all
  become: true
  tasks:

  - name: install apache and php
    apt:
      name:
        - "{{ apache_package }}"
        - "{{ php_package }}"
      state: latest
      update_cache: yes
```

Run *ansible-playbook --ask-become-pass install_apache.yml* and describe the result.

```
isaac@Workstation:~/ansible$ ansible-playbook install apache.yml -K

TASK [install apache2 package] ******************************************************
fatal: [192.168.56.112]: FAILED! => {"msg": "The task includes an option with an undef
ed variable.. 'apache_package' is undefined\n\nThe error appears to be in '/home/isaac
nsible/install_apache.yml': line 5, column 5, but may\nbe elsewhere in the file depenc
g on the exact syntax problem.\n\nThe offending line appears to be:\n\n\n  - name: ins
ll apache2 package\n    ^ here\n"}

PLAY RECAP ******************************************************************************
192.168.56.112             : ok=1    changed=0    unreachable=0    failed=1    skippec
    rescued=0    ignored=0
```

4. Unfortunately, task 2.3 was not successful. It's because we need to change something in the inventory file so that the variables we declared will be in place. Edit the *inventory* file and follow the below configuration:

```
192.168.56.120 apache_package=apache2 php_package=libapache2-mod-php
192.168.56.121 apache_package=apache2 php_package=libapache2-mod-php
192.168.56.122 apache_package=httpd php_package=php
```

Make sure to save the *inventory* file and exit.

```
playbook(first act).txt   install_apache.yml   inventory.ini ×   ansible.cfg

inventory.ini
1   [Isaac]
2   192.168.56.112 apache_package=apache2 php_package=libapache2-mod-php
```

Finally, we still have one more thing to change in our *install_apache.yml* file. In task 2.3, you may notice that the package is assign as apt, which will not run in CentOS. Replace the *apt* with *package*. Package is a module in ansible that is generic, which is going to use whatever package manager the underlying host or the target server uses. For Ubuntu it will automatically use *apt*, and for CentOS it will automatically use *dnf*. Make sure to save the file and exit. For more details about the ansible package, you may refer to this documentation: [ansible.builtin.package – Generic OS package manager — Ansible Documentation]

Run *ansible-playbook --ask-become-pass install_apache.yml* and describe the result.

```
isaac@Workstation:~/ansible$ ansible-playbook install_apache.yml -K
BECOME password:

PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
[WARNING]: Platform linux on host 192.168.56.112 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [192.168.56.112]

TASK [install apache2 package] *************************************************
ok: [192.168.56.112]

PLAY RECAP *********************************************************************
192.168.56.112             : ok=2    changed=0    unreachable=0    failed=0    skipped=0
    rescued=0    ignored=0
```

Now it works, why? Because we just simply declare the package name on our inventory.ini.

Supplementary Activity:

1. Create a playbook that could do the previous tasks in Red Hat OS.

playbook(first act).txt | install_apache.yml | install_redhat.yml | inventory.ini | ansible.cfg

install_redhat.yml

```yaml
- hosts: all
  become: true
  vars:
    apache_package: httpd
    php_package: php
    apache_service: httpd

  tasks:

    - name: install apache and php packages
      dnf:
        name:
          - "{{ apache_package }}"
          - "{{ php_package }}"
        state: latest
        update_cache: yes

    - name: start and enable httpd service
      service:
        name: "{{ apache_service }}"
        state: started
        enabled: yes

    - name: allow http traffic in firewall
      firewalld:
        service: http
        permanent: yes
        state: enabled
        immediate: yes
```

This is the playbook for Redhat OS with the previous tasks.

[EngrAlegre/CPE212_Alegre]

Reflections:

Answer the following:

1. Why do you think refactoring of playbook codes is important?
   - It ensures that code remains clean, scalable, and maintainable, reducing technical debt and improving performance.
2. When do we use the “when” command in playbook?
   - When we are using a conditionals statement, here on activity we use it on

playbook to ensure that a specific task or set of tasks runs only if the target host's operating system is Ubuntu.